Table 1: Client-Side Encryption Complexity Per Round

Scheme	Packs enc.	Cost / pack	Encryption complexity
FedAvg+CKKS	J	$\mathcal{O}(n \log n)$	$\Theta(d \log n)$
FedPHE	sJ	$\mathcal{O}(n \log n)$	$\Theta(sd \log n)$
ESAFL	$\lceil J/2 \rceil$	$\mathcal{O}(n \log n)$	$\Theta(d \log n)$
MASER	J	$\mathcal{O}(n \log n)$	$\Theta(d \log n)$
SAMK	$J + N$	$\mathcal{O}(n \log n) + \mathcal{O}(\log^3 \mathcal{N})$	$\Theta(d \log n) + \Theta(N \log^3 \mathcal{N})$
ECPFL (ours)	$1{+}k$	$\mathcal{O}(n \log n)$	$\Theta\big((k{+}1)\, n \log n\big) = o(d \log n)$

Table 2: Asymptotic Per-Round Cost and Dropout Robustness

Scheme	Client dec.	Client uplink	Interaction	Exp. re-runs (p)
FedAvg+CKKS	$\Theta(d \log n)$	$\Theta(d \log Q)$	1 up + 1 bcast	$\Theta(1)$
FedPHE	$\Theta(d \log n)$	$\Theta(sd \log Q)$	1 up + 1 bcast	$\Theta(1)$
ESAFL	$\Theta(d \log n)$	$\Theta(d \log Q)$	1 up + 1 bcast	$\Theta\big((1-p)^{-N}\big)$
MASER	$\Theta(d \log n)$	$\Theta(d \log Q)$	1 up + N shares	$\Theta\big((1-p)^{-N}\big)$
SAMK	$\Theta(d + N^2)$	$\Theta\big(d \log Q + \frac{Nd}{n} \log \mathcal{N}\big)$	1 up + 1 bcast	$\Theta(1)$
ECPFL (ours)	$\Theta(n \log n + kn)$	$\Theta\big((k{+}1)n \log Q\big)$	1 up + 1 bcast	$\Theta(1)$ for $\lvert H \rvert \geq \tau$

SAMK client decryption includes an N-point interpolation ($\Theta(N^2)$ naive; $\mathcal{O}(N \log^2 N)$ with fast multipoint evaluation).